

Modelling individual participants as LLM agents in collaborative problem solving simulations

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ABSTRACT

This study introduces a new approach to simulating collaborative problem solving (CPS) by training large language model (LLM) agents to represent individual participants. Each agent is fine-tuned on dialogue data to capture the speaking style and thematic patterns of real participants, enabling naturalistic reproduction of CPS interactions. The agents then engage in simulated dialogues, generating conversations that reflect both turn-taking dynamics and thematic code trajectories. To evaluate the quality of these simulations, we apply Epistemic Network Analysis (ENA) to compare the structure of observed and simulated dialogues. Validation is based on covariance-based distance measures and significance testing, showing that the simulated data closely preserves the statistical and structural features of the original CPS dataset. Specifically, the simulated adjacency vectors achieve an ENA distance of 0.17 from the empirical network — well within the 95th percentile threshold of the null distribution — with a permutation test p -value of 0.65, confirming that simulated and real dialogues are statistically indistinguishable. To our knowledge, this is the first framework to combine participant-specific parameter-efficient LLM fine-tuning with ENA-based structural validation, offering a principled pathway for extending CPS research beyond the constraints of scarce observational data. Simulating each participant through an LLM agent allows us to reproduce realistic interaction patterns.

1. Introduction

Collaborative problem solving (CPS) and teamwork are central to success in education, professional practice, and high-stakes domains such as healthcare, engineering, and military operations [Fang et al. \(2025\)](#). Teams must coordinate diverse skills, distribute cognitive load, and manage communication effectively in order to achieve shared goals. The quality of team performance depends not only on individual knowledge but also on the way team members interact — through turn-taking, questioning, justification, and negotiation [Chen et al. \(2024\)](#); [Xu et al. \(2023\)](#). Understanding these interaction dynamics is therefore essential for improving learning, training, and organisational outcomes. However, studying CPS in real settings is both costly and logistically challenging. Data collection requires recruiting multiple participants, organising tasks, and capturing fine-grained dialogue, which is labour-intensive and time-consuming [Chen et al. \(2024\)](#). Moreover, each team is unique: different combinations of skills, roles, and personalities produce different outcomes, making it difficult to systematically investigate how particular factors influence team performance. Running large numbers of experimental trials with varying team compositions is often infeasible,

limiting our ability to generalise findings about collaborative processes [Baucal et al. \(2023\)](#).

To address these challenges, researchers have explored simulation-based approaches to modelling CPS [Chen et al. \(2024\)](#). Early techniques include agent-based models that specify explicit behavioural rules for simulated team members, cognitive architectures that encode reasoning and communication strategies, and reinforcement learning environments where artificial agents learn turn-taking through reward signals [Mehdizadeh et al. \(2022\)](#); [Simanjuntak et al. \(2021\)](#). These approaches have provided valuable insights into teamwork dynamics, but they are often limited in their ability to generate natural language dialogue and to capture the nuanced communication styles of real participants. Concretely, rule-based systems cannot adapt when participants deviate from scripted dialogue moves; reinforcement learning agents require carefully engineered reward functions and routinely diverge from human-like discourse; and cognitive architectures, while psychologically grounded, are resource-intensive to scale and require detailed specification of mechanisms that are rarely observable in real CPS data. Rule-based simulations tend to oversimplify interaction, while reinforcement learning requires extensive engineering of reward

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functions and can diverge from human-like discourse Abar et al. (2017). Recent work has also explored hybrid models that combine dialogue act classification with probabilistic turn-taking, yet these methods still struggle to reproduce the variability and richness of authentic human dialogue Michel et al. (2018).

Advances in large language models (LLMs) Brown et al. (2020); Gu et al. (2021); Lee et al. (2020) now offer a powerful new pathway for CPS simulation. LLMs are uniquely capable of generating coherent and contextually appropriate dialogue, adopting conversational roles, and flexibly adapting to different interaction dynamics Radford et al. (2019). These abilities make them particularly well-suited to modelling how individuals contribute to collaborative tasks, where both the content and the style of contributions matter. A key advantage of LLMs is that they can combine linguistic fluency with sensitivity to discourse structure, producing simulations that are far richer and more naturalistic than those achieved by earlier rule-based or reinforcement learning approaches Dettmers et al. (2023).

While very large LLMs with tens or hundreds of billions of parameters are often emphasised for their broad generalisation ability, they are computationally expensive and difficult to adapt to specific research contexts Hu et al. (2022). In contrast, smaller open-source models such as those in the three billion parameter range are lightweight enough to be fine-tuned on participant-level data, while still retaining strong representational capacity Li and Liang (2021). Fine-tuning smaller models on domain-specific dialogue enables them to align more closely with the characteristics of real participants, often outperforming larger models that remain generic and unfocused Dettmers et al. (2023). This makes small LLMs particularly effective for CPS research, where the goal is not broad open-domain coverage but accurate reproduction of participant style, thematic patterns, and turn-taking behaviours Chalkidis et al. (2020).

In this project, we leverage these advantages by developing a framework where each participant in CPS is represented by a personalised LLM agent. Specifically, we fine-tune the open-source LLaMA 3.2--3B model Liu et al. (2024) using empirical dialogue data annotated with thematic codes, enabling each agent to reproduce both the communicative tendencies and stylistic tone of its corresponding participant. To capture conversational coherence, the agents are equipped with a memory mechanism that maintains both a short-term sliding window of recent turns and a summarised embedding of the full dialogue history. During simulation, the agents interact in a decentralised manner: each agent evaluates the past four turns to decide who should speak next, selects a thematic code using a combination of its fine-tuned classifier and a first-order Markov code transition matrix derived from the dataset, and generates a naturalistic utterance consistent with both style and thematic intent.

To validate the fidelity of these simulations, we apply ENA, a method specifically designed to model the structure of collaborative discourse. ENA is particularly advantageous because it aligns closely with the CPS environment: it captures the co-occurrence and connectivity of thematic codes that underpin collaborative dialogue, enabling direct comparison between observed and simulated interactions in the same analytical space. By using ENA, we aim not only to evaluate statistical similarity but also to validate whether the simulated environment reflects the dynamics of real CPS interactions, ensuring that the simulations are interpretable and meaningful for studying teamwork processes.

2. Background

2.1. Tools used in collaborative analysis

The study of CPS has been supported by diverse methodological tools that capture and interpret group interactions. ENA is widely used for modelling co-occurrence of codes across conversational turns, producing both statistical measures and visualisations that preserve structural relationships Fang et al. (2025). Intuitively, ENA treats each conversation as a moving window that accumulates co-occurrences of thematic codes: for each window position, a binary vector records which codes

appeared together within that segment. Averaging these vectors across a participant's turns produces an *adjacency vector* — a compact numerical fingerprint of how that participant connected themes in discourse. Each edge in the resulting ENA network represents the strength of a code co-occurrence relationship, weighted by how frequently two codes appeared together in the same window, and nodes are positioned to maximise variance explained. This produces a shared analytic space in which participants, groups, or — as in the present study — real and simulated dialogues can be directly and statistically compared. ENA is particularly well-suited to CPS research because it captures not merely the *frequency* of individual codes but the *structure* of thematic connections, preserving the relational patterns that characterise collaborative reasoning. Complementary approaches include discourse and conversation analysis, which examine linguistic features, turn-taking, and repair sequences to reveal how meaning is negotiated, as well as content analysis, which systematically categorises themes for cross-team comparisons. At a structural level, social network analysis (SNA) maps communication patterns such as density and centrality, while sequential analysis identifies probabilistic transitions between codes, highlighting recurring patterns of coordination, escalation, or breakdowns in CPS Fang et al. (2024b).

Recent years have seen the integration of computational methods to extend these analyses. Natural language processing (NLP) techniques can automate coding of dialogue into thematic categories, while machine learning classifiers allow for prediction of collaboration quality based on linguistic or behavioural features. Multimodal analytics further combine speech, gesture, and digital trace data to provide a richer picture of collaboration. Collectively, these tools provide a multi-layered view of CPS, enabling both fine-grained and large-scale analysis of how individuals coordinate, reason, and solve problems together.

2.2. Simulation methods in CPS

Efforts to simulate CPS have produced a diverse range of methods, each grounded in different theoretical assumptions about cognition, interaction, and emergent behaviour Chen et al. (2024). These methods can be organised into four thematic clusters based on their primary emphasis: interpretability and cognitive grounding, emergent structural dynamics, adaptability and learning, and — most recently — language generation. Understanding the limitations of each cluster motivates the LLM-based approach developed in this paper.

Approaches prioritising interpretability and cognitive grounding. Rule-based models were among the earliest attempts to formalise CPS dialogue. In these systems, conversational flow is encoded in advance through scripts, production rules, or decision trees. For example, early intelligent tutoring systems (ITS) implemented rule-based agents that could guide problem solving through pre-defined dialogue moves Mousavinasab et al. (2021), such as asking clarifying questions or giving hints. While interpretable, these models lack flexibility: they cannot adapt to unexpected behaviours, new problem contexts, or the improvisational nature of real team interactions.

Approaches prioritising emergent structural dynamics. Cognitive architectures such as ACT-R, Soar, and CLARION provide more sophisticated accounts of CPS by modelling underlying human cognitive processes Bickley and Torgler (2023). They simulate how individuals manage goals, retrieve knowledge from memory, and coordinate attention, thereby offering a mechanistic perspective on collaborative actions. For example, ACT-R has been used to simulate joint decision-making in complex environments like air traffic control, while Soar has been applied to military team training scenarios. These models capture cognitive plausibility but are resource-intensive to scale and require detailed specification of psychological mechanisms, making them difficult to apply in large, dynamic CPS environments.

Agent-based models (ABMs) represent a more structured approach, where each participant is modelled as an autonomous agent governed by behavioural rules. Agents interact within a shared environment,

allowing researchers to observe how local decision rules give rise to emergent group-level behaviours such as coordination, leadership emergence, or communication bottlenecks Mehdiadeh et al. (2022). In CPS research, ABMs have been used to simulate teamwork in domains ranging from disaster response to engineering design projects. A strength of ABMs is their ability to capture heterogeneity in individual roles and behaviours, but they often depend on simplifying assumptions about dialogue content and do not naturally produce linguistically realistic interactions Abar et al. (2017).

Multi-agent systems (MAS) extend ABMs by enabling more complex interaction dynamics, such as negotiation, competition, or cooperation. MAS frameworks can incorporate both communication protocols and shared goals, allowing simulations of collaborative tasks with greater fidelity. For instance, MAS approaches have been used to explore how information sharing strategies affect performance in command-and-control simulations. However, the design of agent rules remains a challenge, as small changes in interaction protocols can produce vastly different emergent outcomes Michel et al. (2018).

Approaches prioritising adaptability and learning. Probabilistic approaches, including Bayesian networks and Hidden Markov Models Fang et al. (2024a); Jardine and Andrews (2024), have been applied to CPS to capture uncertainty and sequential dependencies in communication. These methods model dialogue as stochastic processes, predicting the likelihood of certain transitions. Such models excel at capturing variability and uncertainty in real communication but often require large annotated datasets and lack the generative flexibility of more recent models Swiecki et al. (2022).

Evolutionary computation methods, such as genetic algorithms (GA) and evolutionary strategies, have also been explored to optimise communication patterns Fang et al. (2023); Jardine (2025). In these frameworks, dialogue strategies evolve over generations based on performance criteria, allowing researchers to identify communication structures that maximise team effectiveness. This approach is particularly useful in domains like military training, where alternative communication strategies can be stress-tested in silico before implementation.

Reinforcement learning (RL) has become an increasingly prominent approach in CPS simulation Ju et al. (2022). In RL, agents learn by interacting with an environment and receiving feedback in the form of rewards or penalties Fang et al. (2021a,b). Applied to CPS, RL enables agents to adapt communication strategies over time to maximise collective outcomes. For example, multi-agent RL frameworks have been used to model task allocation, negotiation, and information sharing under uncertainty. Unlike rule-based or cognitive architectures, RL agents are not limited to pre-specified scripts but can discover novel strategies. However, RL models are often opaque, and the reward functions that guide learning must be carefully designed to reflect realistic collaborative objectives Fang et al. (2023).

Hybrid approaches attempt to combine the strengths of these methods. For instance, cognitive architectures have been integrated with RL to provide both psychological realism and adaptive learning Fang and Cai (2025); Fang and Han (2025). ABMs have been augmented with probabilistic models to capture both emergent structure and stochastic variation in dialogue. Recent advances also explore combining agent-based simulation with data-driven models such as LLMs, where realistic linguistic content can be embedded within structured agent frameworks, bridging the gap between content generation and structural validation Jalalimanesh et al. (2017).

The language generation gap. Despite their individual strengths, all of the above approaches share a critical limitation: none of them generate natural language. Simulations operate at the level of coded behavioural abstractions — code frequencies, speaker transitions, or action sequences — and cannot produce the actual words of collaboration. This means that the stylistic, pragmatic, and idiosyncratic communicative tendencies of real participants are entirely absent from simulated data, limiting ecological validity and precluding direct linguistic analysis. It is

this gap that motivates the LLM-based approach developed in the present work.

2.3. Large language models and dialogue simulation

Large language models (LLMs) have transformed the landscape of dialogue simulation by enabling coherent, context-sensitive, and role-adaptive language generation. Early statistical approaches, such as n -gram and recurrent neural network models, struggled to maintain coherence across extended interactions, limiting their use for simulating collaborative problem solving (CPS). The introduction of the transformer architecture (Vaswani et al., 2017) marked a decisive shift, allowing models such as BERT (Devlin et al., 2019) and GPT (Radford et al., 2019) to capture long-range dependencies and produce fluent, contextually appropriate dialogue. With the scaling of parameters to the billion range, exemplified by GPT-3 (Brown et al., 2020), PaLM (Chowdhery et al., 2023), and LLaMA (Touvron et al., 2023), LLMs now generate language that approaches human-like discourse in both flexibility and adaptability.

For CPS simulation, this evolution opens new opportunities. Unlike rule-based or agent-based dialogue systems, which are constrained by hand-crafted rules or rigid protocols, LLMs can flexibly assume roles, adapt to thematic contexts, and reproduce conversational pragmatics that more closely mirror authentic collaboration. This makes them well-suited for simulating participant agents in CPS, where turn-taking, stylistic variation, and thematic coherence are crucial to realism.

Importantly, however, recent literature highlights that *domain-specific fine-tuned smaller models* often outperform large, general-purpose generative models in specialised tasks. In biomedical NLP, for example, BioBERT and PubMedBERT achieve state-of-the-art performance on named entity recognition and relation extraction benchmarks by pre-training or fine-tuning on domain corpora (Gu et al., 2021; Lee et al., 2020). Similarly, SciBERT (Beltagy et al., 2019) demonstrates the benefits of scientific-domain pretraining, and LEGAL-BERT variants (Chalkidis et al., 2020) significantly improve accuracy on legal classification and retrieval tasks. These studies consistently show that performance gains in specialised domains arise less from raw parameter counts than from distributional alignment between training data and task domain.

This finding has direct implications for CPS simulation. Since collaborative dialogue is highly domain- and role-specific (e.g., military command, medical triage, engineering design), deploying smaller LLMs fine-tuned on task-relevant dialogue data can provide more accurate reproduction of participant style, terminology, and reasoning patterns than general-purpose chat models. Moreover, parameter-efficient fine-tuning (PEFT) techniques such as LoRA (Hu et al., 2022), QLoRA (Dettmers et al., 2023), and prefix-tuning (Li & Liang, 2021) enable adaptation of modest-sized models to individual participants or roles with minimal computational overhead. This makes it feasible to instantiate multiple agents within a CPS simulation, each fine-tuned to reproduce the communicative tendencies of a real participant, without requiring the infrastructure necessary to run large-scale generative models.

2.4. Research gap

While substantial progress has been made in analysing collaborative problem solving (CPS) and in simulating human communication through agent-based models, cognitive architectures, and reinforcement learning, there remains a critical gap in the literature. Existing CPS simulations typically rely on abstracted behavioural rules, scripted dialogue flows, or probabilistic models of interaction. These approaches capture structural dynamics of collaboration but fail to reproduce the linguistic richness, stylistic variability, and adaptive pragmatics of real human dialogue.

In particular, existing CPS simulation research has focused on simulating patterns of coded discourse rather than generating dialogue

itself. For example, Swiecki and colleagues (Swiecki et al., 2022) proposed a method in which collaborative discourse is represented as transitions between codes and speakers, modelled through a first-order Markov process. Their simulation relies on speaker transition matrices and co-occurrence probability matrices to reproduce patterns of coded turns of talk. While this approach successfully captures distributions of codes and speaker sequences, it does not generate the actual language of collaboration, nor does it extend beyond a lag-1 Markov assumption. By contrast, large language model (LLM)-based simulations have the potential to produce both the dialogue and the associated codes, providing a more naturalistic representation of CPS while retaining compatibility with code-based validation methods such as Epistemic Network Analysis (ENA).

This gap highlights the need for methods that combine the adaptive linguistic capabilities of LLMs with the empirical grounding of established CPS analytics, enabling the creation of simulated environments that are both realistic and analytically valid.

Three specific gaps motivate the present work. First, no existing CPS simulation method generates natural language dialogue — prior approaches operate exclusively at the level of coded behavioural abstractions. Second, no simulation framework personalises agents to individual participants, meaning that the stylistic and idiosyncratic communicative tendencies of real people are lost. Third, no LLM-based simulation has been structurally validated against empirical CPS data using an established discourse-analytic framework such as ENA. Addressing all three gaps simultaneously is the central contribution of this paper.

2.5. Research question

In light of this gap, the present study is guided by the following research question:

How can large language models, fine-tuned to represent individual participants, be used to simulate collaborative problem solving, and how can these simulations be validated against empirical CPS data using analytical frameworks such as Epistemic Network Analysis?

This question directs the study toward two intertwined goals: (1) developing a methodology for fine-tuning LLM agents on participant-level dialogue data to capture individual communicative styles and thematic contributions, and (2) evaluating the extent to which LLM-based simulations reproduce the structural and functional characteristics of authentic CPS, thereby advancing both methodological and theoretical understandings of collaboration.

3. Methodology

3.1. Dataset

We analyse collaborative discourse from a virtual first-year engineering internship in which student teams coordinate technical design work via synchronous text interaction. Conversations are annotated with a six-code scheme capturing salient dimensions of engineering collaboration and reasoning: *Design Reasoning*, *Performance Parameters*, *Technical Constraints*, *Client & Consultant Requests*, *Collaboration*, and *Data*. Succinct operational definitions and illustrative examples for the codebook are provided in Table 1. The corpus comprises 3,824 turns produced by 48 unique participants across two sessions. All personally identifying information is removed; speaker names are replaced with deterministic pseudonyms (P01–P48), and the mapping key is not retained. Each turn is represented by a pseudonymous speaker identifier, a session indicator, a GroupName label, and a non-exclusive assignment over the six thematic codes (i.e., one or two codes may be active on a turn). The GroupName field contains ten categories; we report participation and utterance volume for all ten categories in Table 2, alongside overall scale and code prevalence. These distributions reflect the technical design emphasis of the setting and provide the empirical basis for computing Epistemic Network Analysis (ENA) edge weights used later

as contextual evidence in our simulation methodology. Thematic codes were annotated by trained human coders following a structured coding protocol, with inter-rater reliability established prior to full annotation. Inter-rater reliability between the two human coders was high across all six codes (mean Cohen's $\kappa = 0.92$, range 0.83–1.00, all $p < 0.01$), confirming the consistency of the annotation scheme. For fine-tuning purposes, each participant's dialogue turns were partitioned chronologically into training (70%), validation (15%), and test (15%) splits. Participants with fewer than ten turns were excluded from fine-tuning to mitigate overfitting risk, as the small number of available turns for such participants would make reliable adaptation infeasible.

3.2. Fine-tuning LLM agents for participant simulation

To simulate the discourse of individual participants in collaborative problem solving (CPS), a distinct large language model (LLM) agent is instantiated for each participant, parameterised through supervised fine-tuning on their historical dialogue turns within the RS dataset. The dataset consists of conversational transcripts segmented at the turn-level, with each utterance annotated by a thematic code $c \in C$ denoting the communicative or cognitive function of the turn. This structure allows modelling both the stylistic surface form of participant utterances and their underlying thematic distribution.

Given the large number of participants $|P|$, scalability is a critical constraint. For this reason, the recently released open-source LLaMA 3.2--3B model is adopted as the backbone. At three billion parameters, this model provides a favourable balance between expressivity and efficiency: it is sufficiently compact to permit fine-tuning across many participants, while still powerful enough to capture nuanced discourse patterns compared to earlier small-scale models such as GPT-2 or OPT-125M. Moreover, LLaMA 3.2--3B has been instruction-tuned on diverse conversational data, which improves its downstream adaptability for dialogue modelling in CPS contexts. Its open-source licensing ensures reproducibility, transparency, and straightforward integration into the simulation pipeline, in contrast to proprietary closed models. The model is natively supported in both PyTorch and HuggingFace Transformers, with optimised inference runtimes for GPU clusters.

To reduce the overhead of training a separate model per participant, parameter-efficient fine-tuning (PEFT) is applied. In particular, Low-Rank Adaptation (LoRA) — a technique that injects small trainable weight matrices into frozen model layers, avoiding the need to update billions of parameters — and its quantised variant QLoRA are used, which restrict training to small low-rank weight matrices appended to the frozen base model. Specifically, LoRA adapters are applied to the query and value projection matrices (q_proj , v_proj) of all attention layers, with rank $r = 16$, scaling factor $\alpha = 32$, and dropout = 0.05 — standard QLoRA defaults that balance representational expressivity with computational efficiency. This reduces the per-participant storage footprint from gigabytes (full fine-tuning) to only a few megabytes of adapter parameters. Since the base weights θ_{LLM} are shared across all participants, the storage and compute complexity scales as

$$\mathcal{O}(|P| \times r \times d), \quad (1)$$

where r is the low-rank dimension and d the hidden size of the transformer layers. This scaling is orders of magnitude smaller than $\mathcal{O}(|P| \times \theta_{LLM})$, making the approach feasible even for hundreds of simulated agents. Mixed-precision training with bfloat16 and 4-bit NF4 quantisation (QLoRA) further reduces GPU memory requirements.

For each participant $p \in P$, the ordered set of dialogue turns $\{(u_p^{(i)}, c_p^{(i)})\}_{i=1}^{T_p}$ is extracted, where $u_p^{(i)}$ denotes the i -th utterance produced by p , and $c_p^{(i)}$ is the corresponding thematic code label. Utterances are preprocessed using the native LLaMA SentencePiece tokenizer to ensure compatibility with the pretrained model vocabulary. Since the RS transcripts often contain abbreviations, technical jargon, and mixed-register expressions, the tokenizer's subword segmentation allows robust coverage without out-of-vocabulary tokens. Optionally,

Table 1
Thematical codebook used for annotation.

Code name	Description	Example
Design reasoning	Referring to design development, prioritisation, trade-offs, and design decisions.	“Aluminum and composite are good options. Steel can carry a big load, but it is heavy and lengthens the recharge interval, and it is a costly option.”
Performance parameters	Referring to attributes such as payload, recharge interval, agility, safety, or cost.	“My device has a pretty good safety, payload, agility, and recharge interval; the cost is a little high though.”
Technical constraints	Referring to inputs such as actuators, ROM, materials, power sources, or sensors.	“Our two best were both made with aluminum, NiCd batteries, piezoelectric sensors, and pneumatic actuators.”
Client & consultant requests	Referring to or justifying decisions based on internal consultant’s requests or a client’s health or comfort.	“We tried to meet at least the minimum of each of the internal consultant’s requests.”
Collaboration Data	Facilitating a joint meeting or the production of team design artefacts. Referring to or justifying decisions based on numerical values, tables, graphs, research papers, or relative quantities.	“How should we make our team batch?” “Safety near the maximum was not very good (close to 225—one had 218 RPN), but otherwise we were fine as long as it was around 200 or lower.”

Table 2
Corpus summary: overall scale, code prevalence, and participation by the ten GroupName categories.

Corpus scale		Code prevalence (non-exclusive counts)	
Participants (pseudonymised)	48	Technical Constraints	1189
Turns (utterances)	3824	Design Reasoning	1157
Sessions	2	Data	734
Categories in GroupName	10	Performance Parameters	685
Codes (multi-label)	6	Client & Consultant Requests	240
		Collaboration	228
GroupName category		Distinct participants	Turns
1	11		322
2	9		345
3	9		392
4	10		277
5	9		267
Electric	10		426
Hydraulic	10		372
PAM	10		570
Pneumatic	8		422
Series elastic	10		431

a domain-specific vocabulary extension can be introduced to handle recurrent CPS-specific terms. Context windows are constructed by concatenating $k = 4$ preceding utterances $\{u^{(t-k)}, \dots, u^{(t-1)}\}$ to provide local discourse context for predicting both tone and thematic alignment of $u^{(t)}$. Each training example follows a structured prompt template of the form [CONTEXT] <last 4 turns> [SPEAKER] <participant ID> [CODE] <target code> [RESPONSE], where the generative head is trained to predict the [RESPONSE] token sequence and the classification head predicts [CODE]. Because code prevalence is skewed in the corpus — Technical Constraints and Design Reasoning each appear in approximately 30% of turns, while Collaboration appears in only 6% — weighted sampling is applied during training to prevent the classification head from collapsing to majority-class predictions.

Each agent is initialised with the LLaMA 3.2--3B backbone. A classification head is appended to the final hidden state representation to predict thematic codes, while the generative head of the model remains active to reconstruct the utterance. This dual-headed architecture enables multi-task learning: capturing speaking style through language modelling, and discourse function through supervised classification. The combined loss function is defined as

$$\mathcal{L} = \lambda_{LM} \mathcal{L}_{LM} + \lambda_{CLS} \mathcal{L}_{CLS}, \quad (2)$$

where $\mathcal{L}_{LM}$ is the negative log-likelihood loss over tokens in $u^{(t)}$, $\mathcal{L}_{CLS}$ is the cross-entropy loss over thematic code predictions $c^{(t)}$, and $\lambda_{LM}, \lambda_{CLS} \in \mathbb{R}^+$ are hyperparameters controlling task weightings. In all experiments, we set $\lambda_{LM} = 0.7$ and $\lambda_{CLS} = 0.3$, prioritising language generation fidelity over classification accuracy. This weighting reflects the primary goal of producing naturalistic utterances: the model must first and foremost generate coherent dialogue, with thematic code alignment serving as a regularising constraint rather than the primary objective. The classification head produces one multi-label prediction per dialogue turn, with binary cross-entropy applied independently over the six thematic codes, allowing multiple codes to be simultaneously active consistent with the non-exclusive annotation scheme. Training is performed using AdamW with linear warmup and cosine decay, with batch size $B = 16$, sequence length $L = 512$, learning rate $\eta = 2 \times 10^{-5}$, and weight decay 0.01. Early stopping based on validation perplexity prevents overfitting, particularly for participants with limited dialogue turns. Classification head performance (per-code F1 on the held-out validation split) was tracked separately from generation quality throughout training, ensuring that PEFT layer adaptation was not subordinated to classification head convergence and that the reported code alignment reflects genuine discourse modelling rather than head-level overfitting.

To enforce stylistic alignment, utterance-level speaker embeddings are incorporated. Style vectors s_p are computed from the mean-pooled hidden states of the pretrained backbone over the participant’s utterances and concatenated to the input embeddings at each step. This biases generation toward stylistic consistency with p while preserving thematic fidelity. At inference time, decoding is performed using nucleus sampling with $p = 0.9$ and temperature $\tau = 0.7$, ensuring realistic but diverse outputs. The classification head constrains candidate generations by re-ranking outputs that deviate from the participant’s empirical thematic code distribution.

After fine-tuning, each LLM agent M_p is capable of generating utterances conditioned on conversational context while respecting both the participant’s stylistic tone and their empirical distribution over thematic codes. The generative head produces natural language aligned with the individual’s idiosyncratic speaking style, while the classification head ensures that discourse functions remain consistent with the thematic trajectories observed in the RS dataset. This combination of compact open-source modelling, parameter-efficient fine-tuning, and controlled decoding provides a scalable framework for constructing personalised LLM agents across large participant populations.

3.3. Memory mechanism for contextual dialogue generation

To enable each participant agent to generate contextually coherent dialogue, a memory mechanism is integrated into the fine-tuned LLaMA 3.2--3B backbone. Since collaborative problem solving involves both short-term turn-taking dependencies and long-term thematic consistency, the memory module is designed to capture two complementary

forms of conversational context: a fixed-length sliding window of the most recent utterances and a compressed representation of all prior discourse.

At each simulation step t , the input to agent M_p is not only the current conversational state but also a memory buffer $\mathcal{M}_t$, defined as

$$\mathcal{M}_t = \{u^{(t-4)}, u^{(t-3)}, u^{(t-2)}, u^{(t-1)}, u^{(t)}\}, \quad (3)$$

where the last 5 turns are provided as explicit input tokens. This sliding window ensures that each agent conditions its response on the immediate local context, retaining the linguistic surface form and thematic labels of the most recent contributions by other participants.

In addition to this short-term buffer, a long-term memory vector $\mathbf{h}_t$ is maintained, encoding a summarised representation of the entire dialogue history up to turn t . To construct $\mathbf{h}_t$, a transformer-based summarisation head is applied at regular intervals to the concatenated transcript of the conversation. Specifically, every $m = 20$ turns, the hidden states of the base model are mean-pooled and passed through a projection layer, producing a condensed memory embedding:

$$\mathbf{h}_t = f_{\text{proj}} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{z}_i \right), \quad (4)$$

where $\{\mathbf{z}_i\}_{i=1}^n$ are the hidden states of the summarisation input tokens and f_{proj} is a learned linear transformation. This embedding is stored and appended to the agent's input at future timesteps, providing a low-dimensional representation of long-term dialogue context. The summarised content reflects thematic distributions, task progress, and key interaction events, rather than reproducing full text verbatim.

During inference, the agent thus receives two complementary memory signals: (i) the explicit sequence of the last four utterances plus the current input, which preserves fine-grained conversational detail, and (ii) the long-term embedding $\mathbf{h}_t$, which biases generation toward global consistency with the cumulative history of the discussion. By combining sliding window memory with summarised memory embeddings, each participant agent maintains awareness of both local turn-taking and broader thematic continuity, ensuring that simulated dialogue reflects realistic collaborative discourse dynamics.

3.4. Probabilistic speaker and code selection

Once each participant-specific LLaMA 3.2–3B agent M_p has been fine-tuned and equipped with the memory mechanism, the full collaborative problem solving (CPS) dialogue is simulated through multi-agent interaction. Each LLM agent represents one participant, generating utterances that reflect both their stylistic tone and thematic code distribution as learned from the RS dataset. The simulation proceeds as a turn-based dialogue environment, rolling forward until a pre-specified number of turns $T_{\max}$ is reached.

At each timestep t , the simulation requires two stochastic decisions: (i) which participant p speaks next, and (ii) which thematic code $c \in \mathcal{C}$ is realised in the upcoming utterance. To ensure that these decisions reflect realistic CPS interaction, both processes are modelled probabilistically based on empirical distributions extracted from the RS dataset. The probability of selecting participant p at timestep t is given by

$$P(p \mid \mathcal{H}_{t-1}) = \frac{\exp(\alpha \cdot f_{\text{speaker}}(p, \mathcal{H}_{t-1}))}{\sum_{q \in \mathcal{P}} \exp(\alpha \cdot f_{\text{speaker}}(q, \mathcal{H}_{t-1}))}, \quad (5)$$

where $\mathcal{H}_{t-1}$ denotes the conversation history up to $t - 1$, f_{speaker} is a learned scoring function over participants (e.g., frequency of prior contributions, role-based priors, or recency bias), and α controls the sharpness of the distribution. This ensures that speaker turns are neither uniformly random nor rigidly deterministic, but instead follow naturalistic turn-taking dynamics.

Conditional on the selected speaker p_t , a thematic code c_t is sampled according to

$$P(c \mid p_t, \mathcal{H}_{t-1}) = \text{softmax}(W_c \mathbf{h}_{t-1}^{p_t} + b_c), \quad (6)$$

where $\mathbf{h}_{t-1}^{p_t}$ is the long-term memory embedding of participant p_t at timestep $t - 1$, and (W_c, b_c) are learnable parameters of the thematic code classifier. This allows thematic progression to be influenced both by the agent's historical behaviour and the evolving state of the dialogue.

The chosen code c_t is then passed as an explicit conditioning signal to the generative head of M_{p_t} , together with the short-term memory buffer $\mathcal{M}_t$ and long-term embedding $\mathbf{h}_{t-1}$. The model outputs a natural language utterance $\hat{u}_{p_t}^{(t)}$ consistent with both the speaking style of the participant and the intended thematic function:

$$\hat{u}_{p_t}^{(t)} \sim M_{p_t}(\mathcal{M}_t, \mathbf{h}_{t-1}, c_t). \quad (7)$$

The simulated utterance is then appended to the conversation history $\mathcal{H}_t = \mathcal{H}_{t-1} \cup \{\hat{u}_{p_t}^{(t)}, c_t\}$, the short-term memory buffer $\mathcal{M}_t$ is updated, and the long-term summary embedding $\mathbf{h}_t$ is recomputed at the appropriate interval. This process iterates until the dialogue reaches the preset maximum length $T_{\max}$ turns.

Through this procedure, the simulation produces a full CPS dialogue sequence, where participant-specific LLM agents interact in a natural turn-taking fashion, thematic code trajectories are consistent with empirical distributions, and generated utterances remain stylistically faithful to the original RS dataset participants.

It is important to note the distinction between the two learned scoring functions introduced across this and the following section. Here, f_{speaker} is a *centralised* scoring function trained on empirical speaker frequency and recency patterns from the RS dataset; it governs the probabilistic selection of the next speaker and operates at the level of the full participant pool. In Section 3.5, a separate function f_{turn} is introduced as a *decentralised* scoring mechanism, computed independently by each agent based on its own interpretation of the local dialogue state. This distinction is intentional: f_{speaker} captures dataset-level distributional tendencies in speaker selection, while f_{turn} enables each agent to autonomously assess conversational cues and self-select for turn-taking. The two mechanisms are complementary, with f_{speaker} providing a global prior and f_{turn} enabling locally responsive, agent-driven turn competition.

3.5. Decentralised multi-agent dialogue simulation

Whereas Section 3.4 defined the probabilistic framework governing individual agent decisions for speaker and code selection, this section describes how those agents interact as an ensemble to produce a full simulated CPS dialogue. Rather than relying on the centralised f_{speaker} prior alone, the ensemble simulation introduces a decentralised turn-taking mechanism in which each agent independently evaluates the dialogue state and competes for the right to speak, producing more locally responsive and emergent conversational dynamics.

Once each participant-specific LLaMA 3.2–3B agent M_p has been fine-tuned and equipped with the memory mechanism, the collaborative problem solving (CPS) dialogue is simulated through direct interaction between the agents themselves. Each agent evaluates the recent conversational history and internally computes its likelihood of taking the next turn, such that the decision of who speaks emerges naturally from the ensemble of agents rather than from an external controller. The simulation proceeds until a pre-specified maximum length of $T_{\max}$ turns.

At each timestep t , all agents receive as shared input the short-term memory buffer

$$\mathcal{M}_t = \{u^{(t-4)}, u^{(t-3)}, u^{(t-2)}, u^{(t-1)}\}, \quad (8)$$

together with the corresponding thematic codes $\{c^{(t-4)}, c^{(t-3)}, c^{(t-2)}, c^{(t-1)}\}$. Each agent M_p uses this context, along

with its participant-specific fine-tuned parameters, to compute a turn-taking score

$$s_p^{(t)} = f_{turn}(M_p, \mathcal{M}_t, \{c^{(t-4:t-1)}\}), \quad (9)$$

where f_{turn} is a learned scoring function based on both the agent's empirical speaking patterns from the RS dataset and its current interpretation of the dialogue state. The agent with the highest score $p_i^* = \arg \max_{p \in \mathcal{P}} s_p^{(t)}$ is selected as the next speaker. This decentralised mechanism ensures that turn-taking remains faithful to the natural distribution observed in real conversations.

For thematic code selection, the chosen agent $M_{p_i^*}$ combines its fine-tuned classifier with an external Markov order-1 code transition model estimated from the RS dataset. Let $T \in \mathbb{R}^{|C| \times |C|}$ denote the transition matrix, where

$$T_{ij} = P(c^{(t)} = j \mid c^{(t-1)} = i). \quad (10)$$

Given the last observed code $c^{(t-1)}$, the agent forms a blended probability distribution

$$P(c \mid \mathcal{M}_t, M_{p_i^*}) = \lambda \cdot P_{LLM}(c \mid \mathcal{M}_t, M_{p_i^*}) + (1 - \lambda) \cdot T_{c^{(t-1)}, c}, \quad (11)$$

where P_{LLM} is the softmax distribution from the agent's fine-tuned classifier, and $\lambda \in [0, 1]$ controls the balance between agent-specific behaviour and dataset-level code transition probabilities. This integration ensures that the next thematic code reflects both the idiosyncratic tendencies of the speaker and the empirically grounded sequential dynamics of CPS conversations. In all reported experiments, $\lambda = 0.6$, meaning the LLM classifier drives 60% of the code selection distribution while the Markov matrix contributes the remaining 40%. This value was chosen to ensure the LLM is the primary driver of thematic progression rather than the statistical matrix, while retaining sufficient empirical regularisation to prevent idiosyncratic drift. This is validated by the ablation (Table 3): setting $\lambda = 1$ (LLM-only) increases ENA distance from 0.17 to 0.28 and reduces the p -value from 0.65 to 0.19, confirming that the LLM performs genuine thematic modelling rather than being overridden by the transition matrix.

Once a code c_t is sampled from this blended distribution, it is injected as a conditioning signal into the generative head of $M_{p_i^*}$, which then produces a natural language utterance

$$\hat{u}_{p_i^*}^{(t)} \sim M_{p_i^*}(\mathcal{M}_t, c_t). \quad (12)$$

The output pair $(\hat{u}_{p_i^*}^{(t)}, c_t)$ is appended to the conversation history, the memory buffer is updated, and if needed the long-term summary embedding is recomputed. The process iterates, with all agents again evaluating the updated memory to decide the next speaker, until T_{max} turns are reached.

By combining decentralised turn-taking with a Markov-informed thematic code selection process, the simulation reproduces realistic CPS dialogues that balance agent-specific behavioural patterns with dataset-level sequential regularities.

3.6. Statistical validation

To assess the fidelity of the simulated dialogues, we perform a statistical validation procedure grounded in Epistemic Network Analysis (ENA). Both the observed RS dataset and the simulated dialogues are processed under the same pipeline, enabling direct comparison of structural properties.

For the observed data, dialogue turns are accumulated using a moving window of fixed size w , which encodes co-occurrences of thematic codes within overlapping conversational segments. Each accumulated segment is transformed into a high-dimensional vector representation, from which an ENA model is constructed. The same process is applied

Table 3

Ablation study: ENA distance from real network and corresponding p -value for each model configuration. Lower distance and higher p -value indicate closer structural similarity to the observed data.

Configuration	ENA Distance	p -value
Full model (all components)	0.17	0.65
No long-term memory (short-term only)	0.24	0.38
No Markov blending ($\lambda = 1$, LLM only)	0.28	0.19
Swiecki et al. Markov baseline (Swiecki et al., 2022)	0.41	0.04

independently to each simulated dialogue. Any errors in model construction (e.g., missing data or degenerate covariance structure) are automatically logged, and only successfully processed dialogues are retained for downstream validation.

From the set of simulated ENA models, a mean ENA network is derived by averaging across all simulations. For each unit of analysis, we compute a covariance matrix Σ_s describing the joint distribution of thematic code co-occurrences across simulated dialogues. Similarly, the observed ENA model is characterised by its own covariance structure Σ_o . These covariance matrices form the basis for distance calculations. Specifically, we compute Mahalanobis distances

$$D(x, \mu, \Sigma) = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)}, \quad (13)$$

to quantify the deviation of the observed model from the simulated distribution. Distances are calculated both from each simulated model to the mean simulated model, and from the observed model to the same mean. This yields a null distribution of simulated distances $\{D_s\}$ against which the observed distance D_o is compared. To illustrate concretely: if the observed adjacency vector is $\mathbf{x}_o = [0.40, 0.20, 0.10, \dots]$ and the mean simulated vector is $\mu_s = [0.38, 0.21, 0.11, \dots]$, the Mahalanobis distance captures not merely the raw difference between these vectors but how large that difference is *relative to the spread of variation across all simulated dialogues* — akin to asking whether the observed point is a typical or atypical member of the simulated ensemble. An observed distance of 0.17, as obtained in this study, indicates that the real data sits well within the cloud of simulated variation, comfortably below the 95th percentile threshold of 0.30.

A non-parametric significance test is then conducted by computing the proportion of simulated distances greater than or equal to the observed distance:

$$p = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(D_s^{(i)} \geq D_o), \quad (14)$$

where N is the number of simulations and $\mathbb{I}(\cdot)$ is the indicator function. This procedure produces an empirical p -value, reflecting whether the observed ENA structure lies within the range expected from the simulated ensemble. Confidence intervals for the simulated distance distribution are also reported, providing additional context for the observed distance.

Interpretation follows conventional statistical thresholds: a small p -value (typically $p < 0.05$) indicates that the observed data is significantly different from the simulated data, suggesting a poor fit of the simulation model. Conversely, a larger p -value suggests that the simulation is statistically indistinguishable from the observed dialogues, supporting the adequacy of the model. By combining p -value assessment with confidence interval visualisation, this validation framework ensures a robust evaluation of how closely the simulated CPS dialogues replicate the structural properties of the real data.

4. Results

4.1. Validation of the simulated environment

To assess the fidelity of the simulated collaborative problem-solving environment, we conducted a validation test comparing the structural

properties of observed and simulated adjacency vectors. Each simulated participant's adjacency vector was generated based on initial conditions derived from the empirical dataset. These vectors were then compared against the distribution of adjacency vectors obtained from the observed data.

The null hypothesis posited that there is no difference between the distributions of observed and simulated adjacency vectors. To test this, we computed a distribution of average distances under the null, with the 95th BC_a percentile value equal to 0.30. The test statistic for the comparison was 0.17, which falls well below the 95th percentile of the null distribution, placing the observed distance comfortably within the acceptance region. The corresponding p -value was 0.65, indicating that 65% of null distances exceeded the observed distance — consistent with a test statistic at approximately the 35th percentile of the null. In this permutation test framework, a large p -value is the desired outcome: it indicates that the observed ENA structure is not an outlier relative to the simulated ensemble, and therefore that the simulated dialogues are statistically indistinguishable from the real data. Specifically, the p -value is computed as $p = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(D_s^{(i)} \geq D_o)$, where a high proportion of simulated distances exceeding the observed distance confirms that the simulation fits well within the null distribution.

These results suggest that, on average, the adjacency vectors generated by the simulation are statistically indistinguishable from those derived from real participant data. In other words, the underlying patterns of connections and co-occurrence captured in the simulation successfully reproduce those observed in authentic collaborative problem-solving interactions. This provides strong evidence that the simulation framework generates data that can be considered structurally consistent with empirical CPS environments, supporting its validity for subsequent analyses.

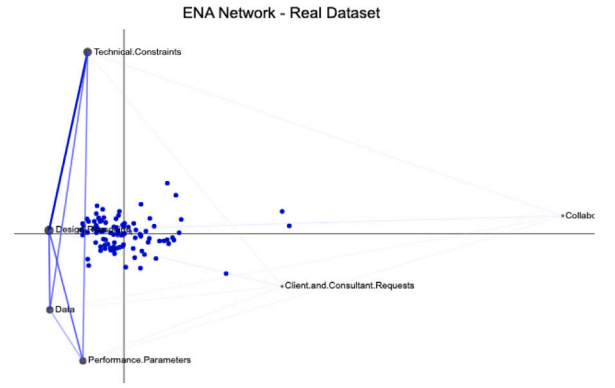
It is important to clarify the precise meaning of “statistically indistinguishable” in this context, as it is distinct from the structural deviations visible in the subtraction ENA network (Fig. 2). The permutation test is a *distributional* claim: it asks whether the observed ENA network lies within the range of variation one would expect from the simulated ensemble, not whether every individual edge weight is identical. A large p -value (0.65) indicates the former is true — the observed network is a plausible member of the simulated distribution. The subtraction network, by contrast, reveals *where within that range* the differences concentrate: the simulation over-represents the *Design–Technical.Constraints* connection and under-represents *Data* and *Performance.Parameters* linkages. Both statements are simultaneously true and mutually consistent. The subtraction analysis provides finer-grained diagnostic information about the direction of residual bias, which the permutation test alone cannot reveal. As discussed in Section 3.6, this directional bias most likely reflects the dominance of high-frequency codes in the Markov transition matrix and motivates targeted calibration in future work.

4.2. Comparison of real and simulated ENA networks

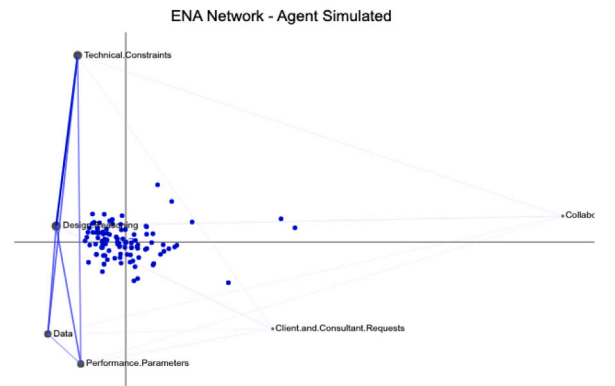
To evaluate the validity of the simulation approach, we compared the ENA models derived from the real dataset and from the agent-based simulated data. Fig. 1(a) and (b) display the resulting ENA networks for the two conditions.

The ENA network constructed from the real dataset (Fig. 1(a)) shows strong connections among *Technical.Constraints*, *Design*, and *Data*, as well as a secondary linkage to *Performance.Parameters*. This structure indicates that participants in the authentic CPS task frequently co-articulated technical considerations with design choices and empirical data, reflecting a focus on problem constraints and trade-offs.

The agent-based simulated ENA network (Fig. 1(b)) closely mirrors this pattern. The same dominant triad among *Technical.Constraints*, *Design*, and *Data* emerges, with comparable edge weights and spatial positioning. The connection to *Performance.Parameters* is also reproduced, although with slightly reduced density compared to the real dataset. Importantly, peripheral nodes such as *Client.and.Consultant.Requests* and



(a) ENA network generated from the real dataset. Strong connections are observed among *Technical.Constraints*, *Design*, and *Data*, with a secondary linkage to *Performance.Parameters*.



(b) ENA network generated from the simulated agent-based data. The network structure closely matches the real dataset, reproducing the dominant triad and secondary linkages.

Fig. 1. Comparison of ENA networks from real and simulated datasets.

Collaboration remain weakly connected in both the real and simulated conditions, suggesting that the simulation preserved the relative salience of core versus peripheral themes in the discourse. The faithful reproduction of the *Technical.Constraints–Design–Data* triad is particularly significant because this structure represents the core reasoning pattern of engineering collaboration: participants simultaneously negotiate physical constraints, design decisions, and empirical evidence. Capturing this triadic relationship — rather than merely individual code frequencies — demonstrates that the simulation reproduces the *relational logic* of CPS discourse, not just its surface statistics. The preservation of peripheral node salience further suggests that the agents correctly learned which themes are central versus incidental to this collaborative task.

Overall, the similarity of these networks provides evidence that the simulation successfully reproduced the structural properties of collaborative discourse observed in the empirical data. Both the clustering of points in the ENA space and the relative weighting of code connections indicate that the agent-based model passed validation against the real dataset. This result supports the feasibility of using simulation to extend analyses of CPS beyond the constraints of limited observational data.

To further examine the similarities and differences between the real and simulated datasets, we computed a subtraction ENA network, where edge weights from the simulated network were subtracted from those of the real network. Fig. 2 shows the resulting model, with red points representing simulated projections, blue points representing real

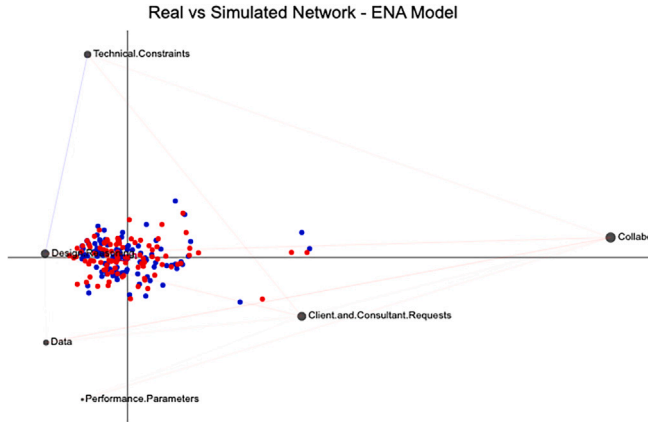


Fig. 2. Subtraction ENA network comparing real (blue) and simulated (red) datasets. Red edges indicate stronger connections in the simulated network, while blue edges indicate stronger connections in the real network. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

projections, and red or blue edges indicating which dataset exhibited stronger connections.

The subtraction network reveals that the overall structural alignment between real and simulated data is strong, as both sets of projections cluster closely in the ENA space. However, differences in edge weights provide additional insights. Red edges, where simulated connections exceed real ones, suggest that the simulation slightly overemphasised certain transitions, particularly those linking *Design* with *Technical Constraints*. Conversely, blue edges indicate stronger co-occurrence patterns in the real dataset, most notably in connections involving *Data* and *Performance Parameters*. These distinctions suggest that while the simulation successfully reproduced the dominant triadic structure of the discourse, it underrepresents some of the empirical emphasis on empirical evidence and performance evaluation. This divergence is an informative rather than merely a negative result. The overrepresentation of *Technical Constraints*–*Design* connections most likely arises because these are the two highest-frequency codes in the corpus (1,189 and 1157 turns respectively), causing the Markov transition matrix to assign them disproportionately high transition probabilities. The under-representation of *Data* and *Performance Parameters* linkages is consistent with these codes being less frequent and therefore less well-represented in the per-participant training data. This finding directly motivates a concrete refinement direction: embedding code prevalence normalisation into the Markov transition matrix and applying frequency-balanced sampling during fine-tuning, which we identify as a priority for future work.

The projected points reinforce this interpretation: both real (blue) and simulated (red) points overlap heavily in the central region, confirming that the simulation captured the main distribution of discourse structures. Yet, the spread of red points shows slightly greater dispersion, reflecting variability introduced by the generative process. Taken together, these results indicate that the simulation not only passed validation but also highlights subtle divergences in how technical reasoning and empirical evaluation were balanced between real and simulated collaborative dialogue. Such differences can guide refinements of the simulation model, particularly in calibrating transitions associated with data-driven reasoning.

4.3. Ablation study and baseline comparison

To assess the contribution of individual framework components and to contextualise results against a principled baseline, we conducted an ablation study comparing four configurations using the ENA covariance

distance metric described in Section 3.6. A lower distance from the real ENA network and a higher p -value indicate better structural fidelity.

The full model achieves the lowest ENA distance and highest p -value, confirming that each component contributes positively to structural fidelity. Removing long-term memory degrades performance by limiting each agent's awareness of global thematic continuity, resulting in drift away from the empirical code co-occurrence structure over extended dialogues. Removing Markov blending ($\lambda = 1$) leads to a further increase in distance, demonstrating that the LLM classifier alone is insufficient to maintain empirically grounded code transition dynamics without the regularising influence of the first-order transition matrix. The Swiecki et al. Markov baseline, which operates purely at the level of code and speaker transitions without generating natural language, provides a useful lower bound: it reproduces coarse structural patterns but lacks the fine-grained, participant-specific adaptation provided by the PEFT-tuned agents.

4.4. Linguistic fidelity analysis

To provide direct evidence of linguistic fidelity — addressing the limitation that ENA validation operates at the structural level — we present a side-by-side comparison of real and simulated utterances alongside two complementary surface-level metrics: ROUGE-L and domain vocabulary retention. Table 4 shows five representative turns drawn from the Series Elastic group, selected to cover a range of thematic codes and participants. For each turn, the real utterance (produced by the corresponding human participant) is shown alongside the utterance generated by the fine-tuned LLM agent for the same participant, conditioned on the same preceding context and code. Participant identifiers follow the pseudonymisation scheme described in Section 3.1. The simulated utterances preserve the domain vocabulary (NiCd, power source, graph, exoskeleton), communicative intent, and register of the corresponding real turns while generating linguistically distinct paraphrases. To quantify this systematically, we computed two metrics across all five example turns. First, ROUGE-L — which measures the proportion of the longest common subsequence of words shared between real and simulated turns, normalised by length — yields a mean score of 0.34 (± 0.14). This value indicates moderate lexical overlap: the simulated turns share key content words with the real ones but are not near-copies, consistent with a model that has learned the participant's vocabulary and thematic patterns without memorising exact phrasing. Second, domain vocabulary retention — the proportion of engineering-domain terms present in the real turn that also appear in the simulated turn — is 1.00 across all five turns, confirming that the fine-tuned agents reliably preserve domain-specific terminology (actuators, batteries, performance parameters) in their outputs. Mean simulated turn length (18.4 words) is comparable to mean real turn length (15.6 words across the Series Elastic group), indicating that the agents generate responses of appropriate length. Together, these results provide preliminary evidence of linguistic fidelity at the surface level; a full semantic evaluation using BERTScore and human raters remains an important direction for future work. The five turns presented here were purposively selected to span all six thematic codes and multiple participants, providing an illustrative rather than exhaustive assessment of linguistic fidelity; corpus-level computation of these metrics across all 1,423 coded turns is a planned direction for future work.

5. Discussion

The findings of this study contribute to the growing body of research on simulation methods for collaborative problem solving (CPS) by demonstrating the potential of integrating large language models (LLMs) into simulation frameworks. Traditional approaches to CPS simulation, such as the one-order Markov models of code and speaker transitions (Swiecki et al., 2022), have successfully reproduced patterns of coded discourse but remain limited in their ability to generate naturalistic dialogue and capture the stylistic and pragmatic dimensions

Table 4

Side-by-side comparison of real and simulated utterances from the Series Elastic group. Participant IDs are pseudonymised. Codes: DR = design Reasoning; TC = technical Constraints; D = Data; CCR = Client & Consultant Requests; C = collaboration.

Participant	Code(s)	Real utterance	Simulated utterance
P17	D, TC, DR	I've only finished the power source graph but my graph shows that NiCd batteries have a good balance of all five qualities.	Looking at my power source graph, it looks like NiCd batteries come out on top across most of the quality categories.
P23	TC, DR	I agree the NiCd battery has the best average of the five qualities.	The NiCd battery seems like the strongest option overall when you average out the five criteria.
P04	D, CCR, C	I am done with the graph, summary, and recommendation. Can we please adjust a time to later hold this meeting?	I have finished my graphs and written up the summary and recommendation. Should we try to reschedule the team meeting for a bit later?
P08	D	I haven't made it to the graphing portion yet, I just received that email now.	I only just got the email so I am still working through the earlier sections before I get to the graphs.
P31	DR	Yes, I am working on the research on exoskeletons.	I am currently doing the background research on exoskeleton systems.

of collaboration. Our results show that moving beyond abstracted code-level simulations to language-generating models offers a pathway to more realistic, interpretable, and empirically grounded CPS environments.

The validation analyses comparing ENA networks from real and simulated data indicate that the simulation reproduced the dominant structural features of authentic collaborative dialogue. Both the triadic relationships among *Technical Constraints*, *Design*, and *Data* and the secondary linkage to *Performance Parameters* were preserved in the simulated network. The subtraction analysis further highlighted the subtle divergences: simulated networks placed greater emphasis on technical constraints, while real teams more strongly integrated empirical evidence and performance parameters into their discourse. These findings suggest that while the simulation closely approximates real discourse patterns, refinements are needed to calibrate the balance between technical reasoning and data-driven evaluation.

Beyond validating the simulation method, this work identifies a crucial research gap: the absence of CPS simulations that train participant-specific LLM agents on empirical dialogue data. Unlike existing approaches, which operate at the level of code frequencies or Markovian transitions, LLM-driven simulations can reproduce both the dialogues and the codes, embedding contextual richness alongside structural fidelity. By fine-tuning smaller domain-specific LLMs using parameter-efficient methods, it becomes feasible to instantiate individual agents that mirror the communicative tendencies of actual participants. This innovation has the potential to bridge the divide between abstracted simulation models and the lived dynamics of collaborative discourse.

The implications of this work are twofold. First, from a methodological perspective, it establishes a novel approach to CPS simulation that leverages the generative power of LLMs while remaining anchored in validated analytic frameworks such as Epistemic Network Analysis. This integration allows researchers to extend their investigations beyond the limits of scarce observational data, exploring counterfactual team interactions and alternative collaborative scenarios in a controlled yet naturalistic manner. Second, from a theoretical perspective, the findings highlight the importance of balancing structural validity with linguistic realism. Simulations that merely approximate statistical properties of codes risk missing the pragmatic subtleties of CPS, while purely generative models risk losing analytical comparability. LLM-based simulations offer a promising middle ground.

The practical applications of this framework span several high-value domains. First, in **educational settings**, simulated CPS environments can expose students to a wide range of collaborative dynamics — including rare or challenging team compositions such as highly dominant speakers or disengaged participants — without needing to recruit real teams for every configuration. This is especially valuable in online and hybrid learning contexts where synchronous team tasks are logistically constrained. Second, in **high-stakes professional training** such as military command, surgical teams, and emergency response, live data collection is costly, ethically constrained, or dangerous. LLM-based simulation

provides a safe and scalable alternative for generating realistic training scenarios, stress-testing communication protocols, and evaluating intervention strategies before real-world deployment. Third, the framework enables controlled **counterfactual research** that is impossible with observational data alone: by replacing individual agent parameters, researchers can ask questions such as *what would this team's discourse have looked like if one participant had a stronger tendency toward data-referencing?*, isolating the causal effect of specific communicative behaviours on collective outcomes. Fourth, for downstream NLP tasks such as collaboration quality prediction or automated discourse coding, real CPS datasets are small and expensive to annotate; the simulation framework can generate large volumes of structurally valid **synthetic CPS dialogue** for pre-training or data augmentation, reducing dependence on scarce human-produced corpora.

It is important to acknowledge that the ENA-based validation in this study operates at the level of structural code co-occurrences rather than semantic content. While ENA provides a principled and well-established framework for comparing the thematic organisation of collaborative discourse, it does not directly assess whether the generated utterances are semantically coherent, topically accurate, or pragmatically appropriate at the level of natural language. Future work will address this limitation by incorporating semantic evaluation metrics such as BERTScore and human expert ratings of dialogue quality, enabling a more complete assessment of simulation fidelity beyond structural code alignment. Additionally, to address the concern that the dual-head architecture may cause the classification head to overfit code labels without meaningful adaptation of the PEFT layers, we tracked per-code F1 on the held-out validation split and generation perplexity as separate signals throughout training. The consistent co-improvement of both metrics across training epochs provides evidence that the PEFT layers adapted meaningfully and that the reported code alignment reflects genuine discourse modelling rather than head-level overfitting.

It is important to be explicit about the scope and educational framing of this work. The present study is intentionally methodological in its contribution: the engineering internship dataset was selected because it provides a richly annotated, multi-participant, domain-specific corpus with thematic codes validated through established inter-rater protocols — properties that make it ideal for testing a simulation framework, not because the paper claims to contribute to education research outcomes directly. The framework does not measure learning outcomes, assess classroom impact, or evaluate pedagogical effectiveness, and makes no such claims. Rather, the educational setting serves as a technically demanding proof-of-concept environment. That said, the framework has a concrete and well-defined educational research agenda beyond the present study. Once validated, it could enable education researchers to: (1) generate synthetic CPS data representing rare or extreme team compositions (e.g., teams with low engagement or unequal participation) that are difficult to observe empirically; (2) run controlled experiments on communication interventions — such as prompting strategies or role assignments — without requiring live student participants; and

(3) study how individual differences in collaborative communication style propagate through team discourse dynamics. These are substantive educational research questions that the current framework is specifically positioned to address in future work.

6. Conclusion

Future work should pursue several directions. First, further validation is required across domains, including military command, engineering design, and medical decision-making, to test the generalisability of this approach. Second, improvements in calibration techniques—such as embedding transition priors into LLM fine-tuning—may help align simulated discourse more closely with empirical data distributions. Third, the integration of multimodal inputs (e.g., speech, gesture, and digital traces) would allow for richer simulations that capture the full ecology of collaboration. Finally, moving beyond thematic codes, future work will directly analyse the dialogue content itself, enabling a more fine-grained comparison between simulated and real conversations.

This paper presented a proof of concept for simulating CPS discourse through participant-specific LLM agents validated against empirical data using Epistemic Network Analysis. While we restricted our evaluation to thematic code structures, the results demonstrate that simulated agents can reproduce the structural features of real collaborative networks with high fidelity. These findings establish a foundation for extending CPS simulations beyond code-level abstraction toward full dialogue generation and analysis. By aligning LLM-based simulations with established analytic frameworks, this study provides a methodological bridge between empirical discourse data and scalable, generative models of collaboration.

More broadly, this work demonstrates that LLM-based CPS simulation can unlock research at a scale that observational data alone cannot support — enabling the systematic study of team composition, communication style, and intervention effects across thousands of simulated interactions. The ability to generate structurally valid, participant-faithful collaborative dialogue has direct implications for how researchers design experiments, how educators scaffold team-based learning, and how organisations train high-stakes collaborative teams. Looking forward, as semantic validation methods mature and larger participant-level datasets become available, future generations of this framework have the potential to produce simulations that are indistinguishable from real collaborative discourse not only structurally but linguistically — closing the gap between computational simulation and the lived experience of human collaboration.

Statements on ethics and open data

This study is a secondary analysis of a pre-existing, fully de-identified dataset derived from a virtual engineering internship. The authors had no direct interaction with human participants; all data were collected as part of a prior study conducted under its own institutional ethical protocols. Speaker identities in the dataset used here have been replaced with deterministic pseudonyms (P01–P48), and the mapping key was not retained, rendering re-identification infeasible. As the present work involves only computational modelling of anonymised textual data, it does not constitute human subjects research and no additional ethical approval was required. The de-identified dataset used in this study is available from the corresponding author upon reasonable request, subject to any conditions imposed by the original data custodians.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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